Improve One Cause

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Locate the defect. Change its cause. Check what used to work.

Repair equality while preserving quiet, validation and order, then fix the candidate before final evaluation.

Keep one habit: Keep the failure → make the targeted edit → rerun prior successes → retain every version.

MAIN LESSON · SECTION 1

Did the repair preserve what already worked?

> Target an evidenced cause and inspect every changed outcome.

Start with a repair that introduces another failure

A repair can fix equality while breaking the quiet requirement. Check the located defect and previously correct behaviour under unchanged requirements, and keep the old evidence.

Recall the inclusive time requirement

The faulty S comparison rejects an activity that takes exactly the available time. Which requirement does this break?

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Your learning goals

- I can **repair** a located **defect** without changing the agreed requirements.
- I can check the **repair**ed case and earlier successful behaviour.
- I can preserve versioned **evidence** and fix a candidate before final evaluation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Test Edge Cases and Fairness. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Build a Paper Prototype. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Find a Real Problem. Give an example and a mistake that the idea helps you avoid.

An improvement should target **evidence** from a failure, not random change.



LEARNING TARGET Version A makes four errors. Version B makes one after lighting, labels and rules all change. What can you conclude, and how would you isolate the effect of lighting?

Learner name:

Readiness for the connected explanation

This week's end goal

By the end, I can **repair** one identified cause and rerun cases that previously passed.

MAIN LESSON · SECTION 2

Keep the activity requirements unchanged during repair

> Use the same catalogue and evidence-linked version labels.

Define repair and regression checks

Finding a failure is useful only if you can explain and check the **repair**. This week you will locate an incorrect comparison, change it in a labelled version, and test both the revealing case and behaviour that worked before.

A **repair** can solve one problem while introducing another, so one newly passing example is not enough.

Read the unchanged activity catalogue

Keep catalogue B version 1 in order: B1 Bookmark drawing, 10 minutes, quiet Yes; B2 Story circle, 8 minutes, quiet No; B3 Pattern puzzle, **12 minutes**, quiet Yes; B4 Tower challenge, 15 minutes, quiet No.

These fictional values assume materials are available. **Input**s are whole minutes from 0 to 30 inclusive and Quiet required Yes or No. No allows either quiet or nonquiet activities.

Preserve the complete contract and reset rules

A candidate fits when its duration is no more than the available minutes AND either quiet is not required or the activity is quiet. Return the first fit with a correct reason.

Valid **input** with no fit gives no match after all rows fail. Invalid or missing **input** needs clarification before search. Each new run starts with validation and a cleared **output**, then the first row if valid.

Suggestions do not book or publish anything. These **written rule**s are not a model fitted from examples.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY The sorter misclassifies worn labels, so the team changes the text-reading rule or adds a safe visual clue while keeping other parts fixed.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

Locate S0’s comparison defect and save S1

> Repair equality without changing quiet, validation, order or stopping.

Inspect the controlled repair cycle

A repair with evidence

LOCATE Keep the failing trace	**CHANGE** Save a new version	**CHECK** Rerun the revealing case
LOCATE Name the broken requirement	**CHANGE** Correct the faulty step	**CHECK** Rerun earlier passing cases
LOCATE Keep input and catalogue labels	**CHANGE** Record what changed	**CHECK** Keep every result and limit

A passing repair test checks the reported defect. Regression tests examine previously correct behaviour.

Connect the failing trace to the broken requirement

S version 0 follows the P1 **process** except that its time check uses strictly less than. On 10 Yes with catalogue B version 1, it reports no match. The expected result is B1 because 10 fits exactly and B1 is quiet.

Compare the saved instruction with the trace: the cause is the exclusion of equality, not a need to change the catalogue or expected answer.

Write the exact change record and preserve the original

- Create S version 1 by replacing only the time comparison with no more than. Keep validation, the quiet condition, catalogue order, advancement and stopping rules.

  Write a **change record** naming S0, S1, the failing 10 Yes case, the comparison changed and why equality belongs in the rule. Retain the old version and failed trace; do not overwrite them with the new result.

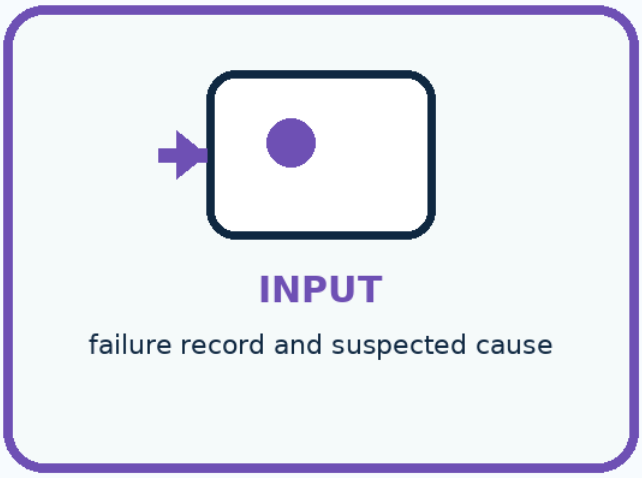
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

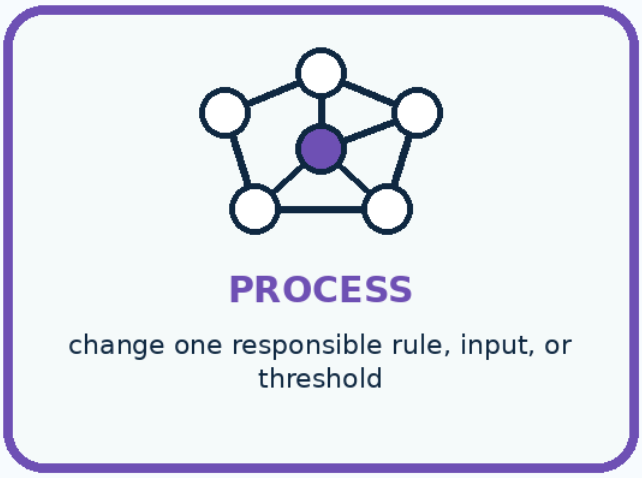
▼ Earlier wording, alternatives and picture transcripts

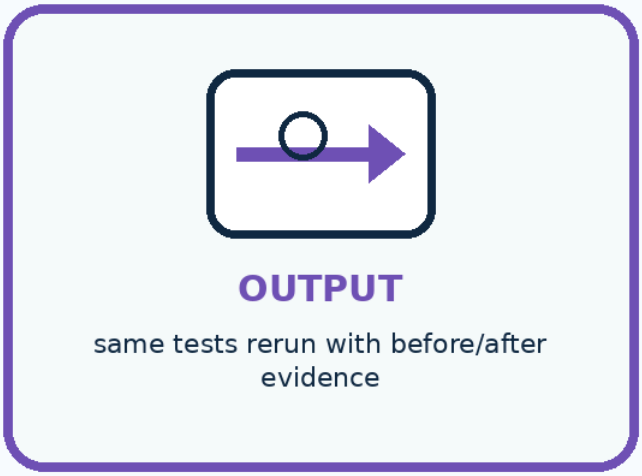
VISUAL **MECHANISM**

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT failure record and suspected cause

PROCESS change one responsible rule, **input**, or threshold

OUTPUT same tests rerun with before/after **evidence**

CORE IDEA An improvement should target **evidence** from a failure, not random change.

UNDER THE HOOD Controlled iteration links an intervention to an observed outcome. Holding test cases and unrelated components constant strengthens causal inference.

Improvement claims must include **regression**s and tradeoffs, not only **repair**ed cases.

Draw a fresh example with all three stages:

A closer explanation

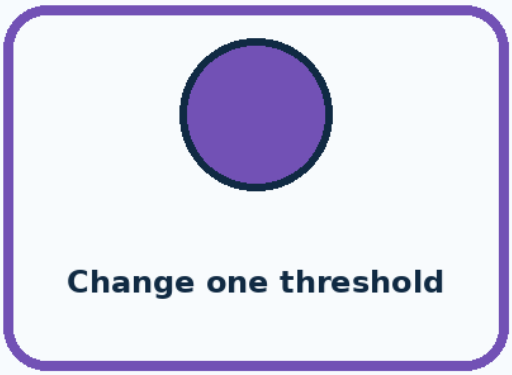
Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare targeted repairs with uncontrolled changes

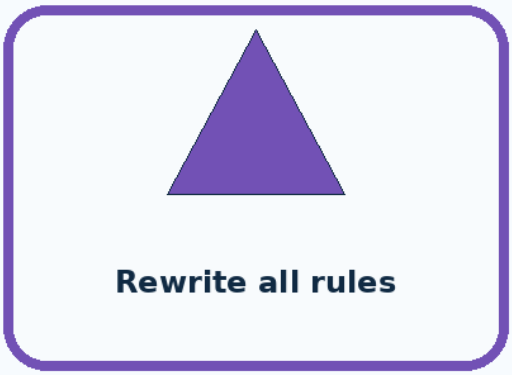
> Review every new failure, not only a higher total pass count.

Pictures paired with their evidence



The case to investigate

CASE 1 — Change one threshold Clue: targeted intervention. Meaning: interpretable.



The case to investigate

CASE 2 — Rewrite all rules Clue: many causes changed. Meaning: weak **evidence**.



The case to investigate

CASE 3 — Same tests Clue: before/after comparable. Meaning: stronger **evidence**.



The case to investigate

CASE 4 — New failure Clue: old case now breaks. Meaning: **regression**.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep the limit in view

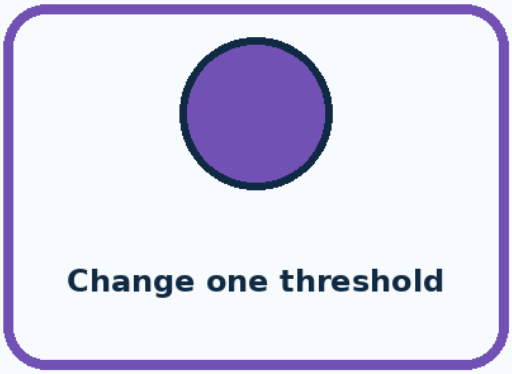
BOUNDARY A higher pass count can hide a serious new failure; review every changed outcome.

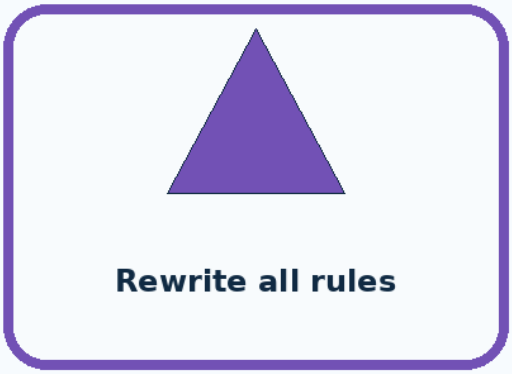
▼ Earlier wording, alternatives and picture transcripts

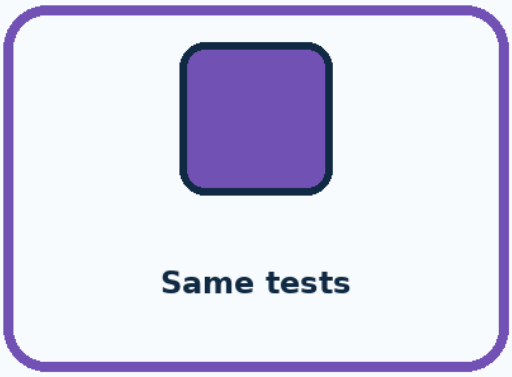
PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Change one threshold Clue: targeted intervention. Meaning: interpretable.

CASE 2 — Rewrite all rules Clue: many causes changed. Meaning: weak **evidence**.

CASE 3 — Same tests Clue: before/after comparable. Meaning: stronger **evidence**.

CASE 4 — New failure Clue: old case now breaks. Meaning: **regression**.

Change one thing in the pictured case

Change: Use U2, which changes < to ≤ and keeps the quiet check.

Prediction: At 8 return A; at 6 return none.

Reason: Equality is **repair**ed while B remains ineligible because it is noisy.

Compare targeted repairs with introduced regressions

> Start with the main activity S/T versions, then compare the separate two-activity U example.

Inspect T1 against the main activity requirements

Consider a different proposed **repair**, T version 1. It accepts equal times but accidentally removes the quiet check. It fixes 10 Yes, yet now returns B2 for 9 Yes.

That second request correctly returned no match under S0 because B2 is not quiet. Previously correct behaviour has become wrong: this is a **regression**.

Case with catalogue B version 1	Required result	Outputs from the stated instructions
10 Yes; equality repair	B1 with a valid reason	S0 no match; T1 B1; S1 B1
9 Yes; quiet regression check	No match with a valid reason	S0 no match; T1 B2; S1 no match
8 No; another equality case	B2 with a valid reason	S0 no match; T1 B2; S1 B2
6 No; exhaustion check	No match with a valid reason	S0 no match; T1 no match; S1 no match

These **output**s are worked **prediction**s from the described versions. Your own test record must contain the actual runs and their reasons. T1 should not replace the candidate merely because it passes the first row.

For S1, check the revealing equality cases and rerun all four routine cases from Week 33, then validation and the labelled catalogue variations from Week 34.

A full pass still needs both the required **output** and a correct explanation.

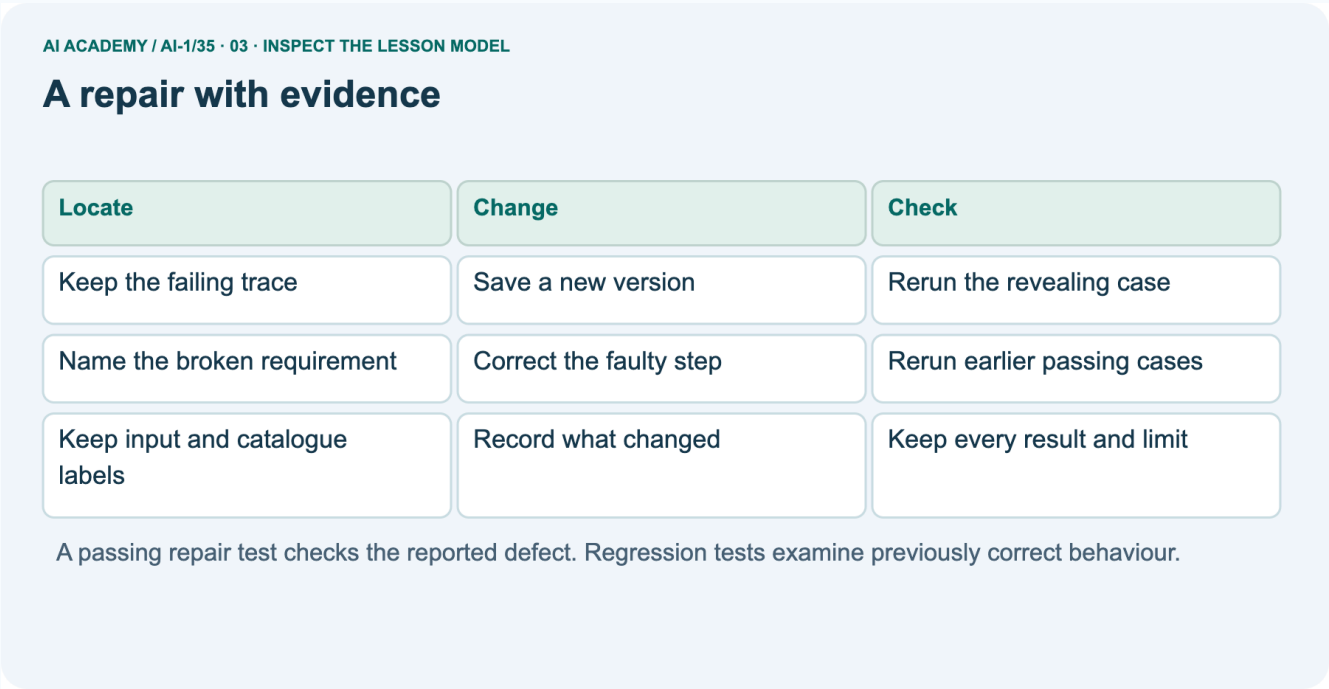
Read the two activities and faulty revisions

A fictional quiet-activity chooser has a boundary bug and a proposed **repair**.

A: quiet Yes, 8 minutes; B: quiet No, 5 minutes. Request quiet required Yes.

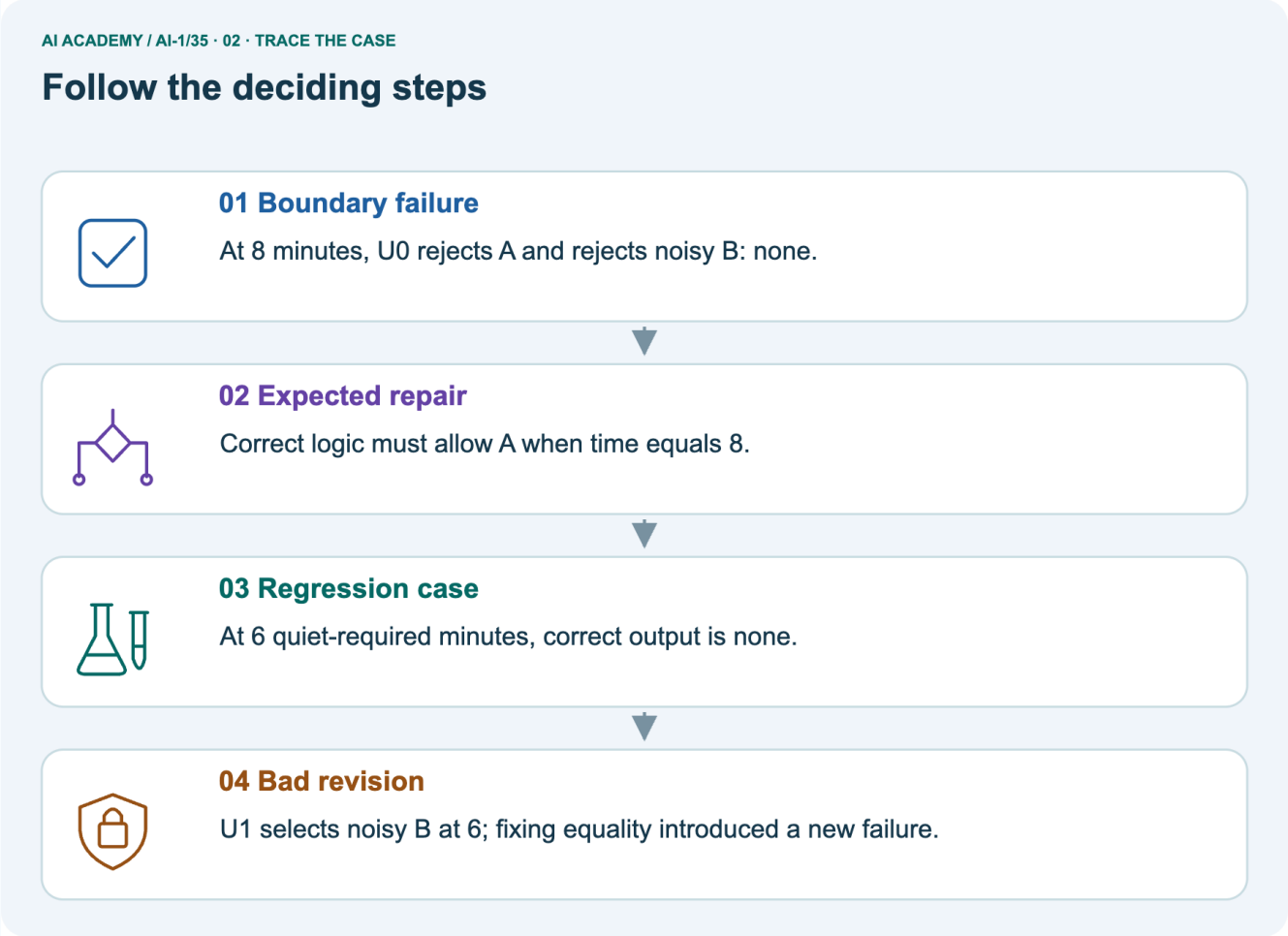
U0 accepts only duration<available time. U1 fixes ≤ but accidentally removes the quiet requirement.

Inspect the labelled repair comparison



Follow the labelled picture with the matching explanation. Open the vector diagram

Trace the equality fix and quiet regression



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Boundary failure	At 8 minutes, U0 rejects A and rejects noisy B: none.
2. Expected repair	Correct logic must allow A when time equals 8.
3. Regression case	At 6 quiet-required minutes, correct output is none.
4. Bad revision	U1 selects noisy B at 6; fixing equality introduced a new failure.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Explain why merely selecting something is insufficient

- A **repair** must preserve the full contract, not just make one failing example pass. The quiet condition and duration comparison are separate requirements.

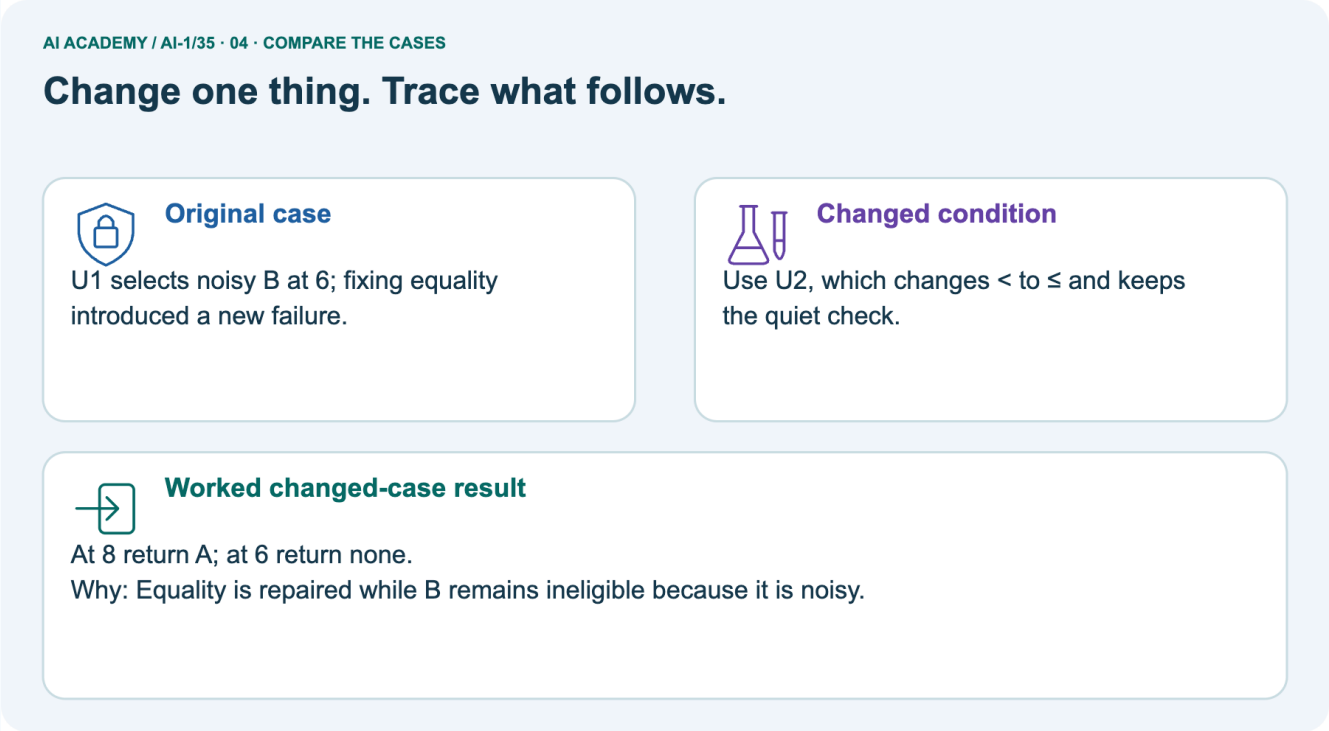
  Changing only the inequality targets the recorded cause; removing the quiet check changes another behavior. Save previously passing cases to reveal such **regression**s.

- Read the labelled picture: Explain why “our **repair**ed version now picks something” is not enough to show success at 6.

Predict the targeted U2 repair

- Before opening the U2 picture, replace < with ≤ while keeping the quiet condition in the two-activity A/B example. Predict outputs at eight and six minutes with quiet required Yes. Which former success must remain correct?

▼ Check the preserved quiet condition



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep the two-example limit

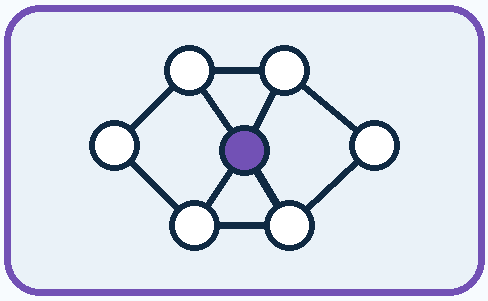
Two examples establish only these outcomes; further cases are needed before generalising the **repair**.

▼ Earlier wording, alternatives and picture transcripts

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE The sorter misclassifies worn labels, so the team changes the text-reading rule or adds a safe visual clue while keeping other parts fixed.

STEP 1 — OBSERVE failure record and suspected cause

STEP 2 — TRACE change one responsible rule, **input**, or threshold

STEP 3 — CONCLUDE same tests rerun with before/after **evidence**

MODEL REASONING Name the **evidence**, suspected cause, single change, and predicted effect before editing. Rerun the full fixed suite, compare case by case, and keep the change only if benefits outweigh new failures.

DO NOT OVERCLAIM A higher pass count can hide a serious new failure; review every changed outcome.

What new **evidence** could change this answer?

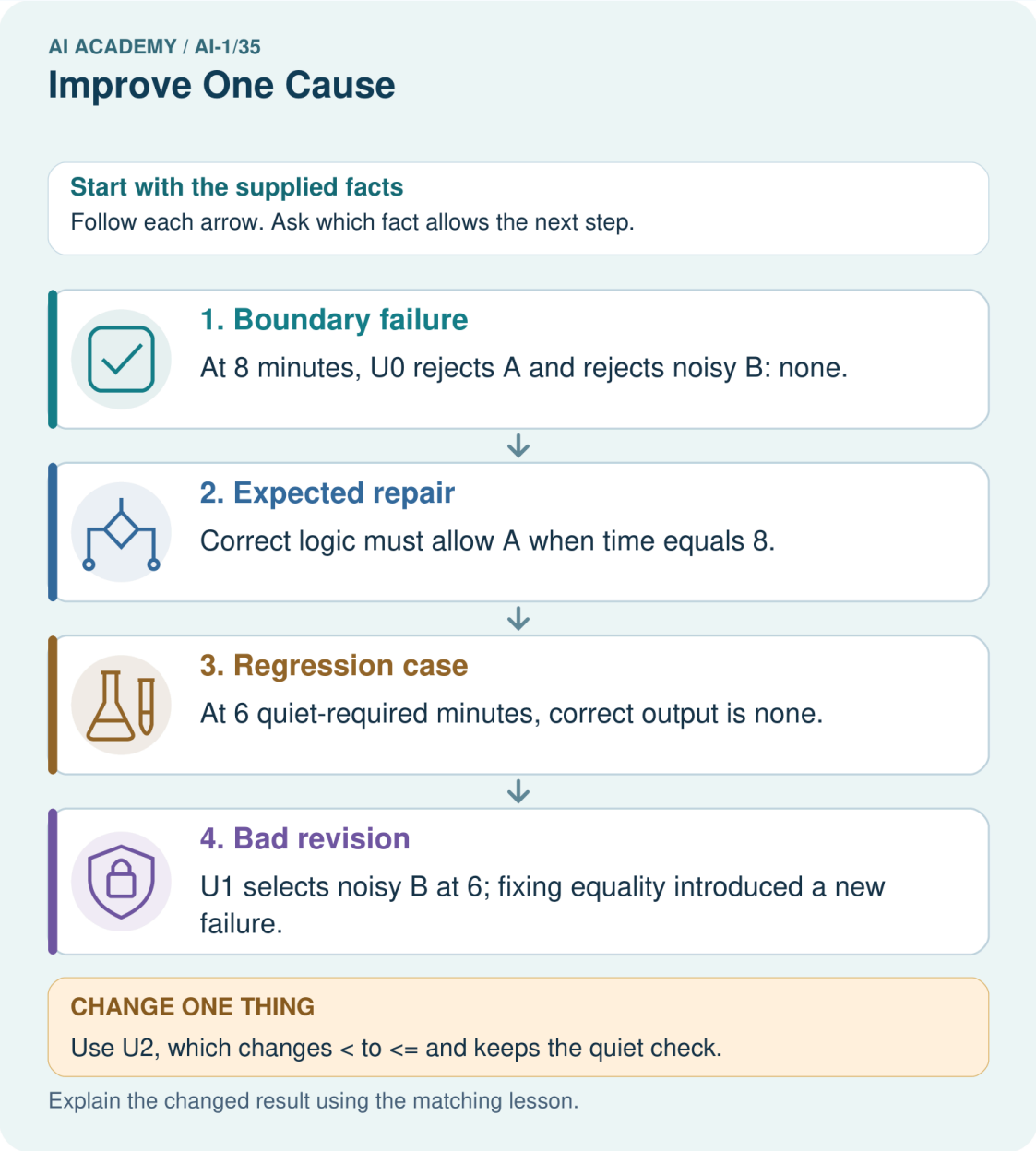
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Improve One Cause

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing: Use U2, which changes < to ≤ and keeps the quiet check. Predict the result before checking the explanation. What remains the same?

MAIN LESSON · SECTION 7

Keep a justified candidate for final evaluation

> If no defect is evidenced, retain the tested version instead of manufacturing a change.

Choose repair from actual failure evidence

If your own P1 already passes the development set, do not introduce a **defect** just to claim an improvement.

Use the supplied S example to practise **repair**, and report that no change was needed in your saved version for the tested cases. If your version fails, retain the **evidence**, locate the cause and label the next **repair**.

A new label alone does not prove a better version.

Save the candidate before reserved cases are opened

Save a candidate with its complete instructions, required catalogue data, **input** rules, change history and development results. State unresolved failures and limitations. Keep this version unchanged during the final evaluation.

Do not inspect reserved final cases to decide which **repair** to make.

If you revise after seeing final results, retain those results as belonging to the old version and treat the seen cases as development **evidence**; they are no longer an unseen evaluation for the revision.

Repair the weak change-history example

> **A team fixes 10 Yes, checks only that case and deletes S0's failed trace. Explain two weaknesses in its evidence and give a better next action. Why should the candidate be fixed before reserved final cases are revealed?**

Repair both broad claims

- CLAIM A Any change after a failure counts as improvement.
- CLAIM B **Retest** only the failed case to save time.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

▼ Earlier wording, alternatives and picture transcripts

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Change one threshold	targeted intervention	interpretable
Rewrite all rules	many causes changed	weak evidence
Same tests	before/after comparable	stronger evidence
New failure	old case now breaks	regression

Repair two tempting claims

Repair:

Repair:

A closer explanation

MAIN LESSON · SECTION 8

Practise: rerun the repair and earlier successful paths

> Keep worked predictions separate from actual observed runs.

Plan routine, equality and validation checks

Write expectations before running S1. Use B version 1 for 11 Yes, 9 No, 6 No and 9 Yes; include the revealing 10 Yes and 8 No.

Add 0 Yes, 30 Yes, 31 Yes, blank minutes with Yes, 10.5 Yes and 10 Maybe to check valid endpoints and invalid fields. Keep cases and results separated by version, and compare with the manual baseline under the same conditions.

Retain the labelled later-row catalogue variations

Retain the catalogue variation tests: B version 2 changes only B1 time to 20, giving B3 for 12 Yes.

B version 3 has times 20, 18, 17 and 15 for B1 to B4, with the original quiet labels and order; it gives B4 for 15 No and no match for 15 Yes.

These cases check later successful paths and the quiet condition under changed data. Label each test with the catalogue it actually uses.

Distinguish regression from an existing defect

> **Explain why T1′s result on 9 Yes is a regression but S0′s result on 8 No is an existing defect. State the required result in each case and the instruction that explains the mistake. If S1 passes both, what further evidence should you keep before making a broader claim?**

Operate the fixed-version set in W2

Use workbook W2 for the connected example “Improving and checking **regression**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge the one-example guarantee

- Use the worked case for Improving and checking **regression**s. Proposed explanation: “One **repair**ed example proves improvement everywhere”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your **repair** if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Return to the quiet-regression example

Explain why “our **repair**ed version now picks something” is not enough to show success at 6.

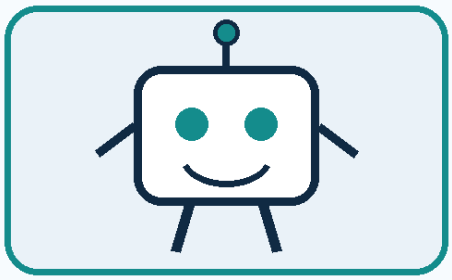
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: An improvement should target **evidence** from a failure, not random change.

Use a fresh example to explain **root cause**.

Trace a new **input** → **process** → **output** example.

Find and **repair** the error in: **Retest** only the failed case to save time.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Practise with less help: Check the repair and earlier behaviour

Check your reasoning

Explain the picture

> **My explanation using a pictured step and the given facts**

MAIN LESSON · SECTION 9

Investigate an explicit one-cause repair and regression set

> Use the supplied defect rather than inventing a failure in an already-correct project.

The source defect, expected result and controlled edit

One-cause paper repair cycle using the supplied main activity example. S0 is P1 except its time condition is duration<available minutes. Save S0, its (10,Yes) no-match failure and the expected B1 result. Create S1 changing only < to ≤, leaving all other validation, quiet, order and stopping rules unchanged. If your own P1 has no evidenced defect, use this supplied example for repair practice and keep your actual version unchanged.

Predict, rerun and preserve the complete evidence

1. Before operating S1, write expectations for the original R cases (11,Yes),(9,No),(6,No),(9,Yes), the equality cases (10,Yes),(8,No), the validation/endpoints (0,Yes),(30,Yes),(31,Yes), (blank,Yes),(10.5,Yes),(10,Maybe), and the labelled B version 2/3 variations defined in the lesson. Run the manual baseline and S1 under identical conditions, record actual results and trace references, and retain all failures. Separate unrun cases. Save the full change history and fixed candidate before the teacher reveals reserved final cases.

2. Keep a normal case and an edge case clearly separated.

3. Record what happened—even when it does not match your **prediction**.

4. Explain the result, one limitation, and the next useful test.

Keep versions, actual traces and evaluation limits

Case	Prediction	Observed result	What it means
Normal			
Edge			

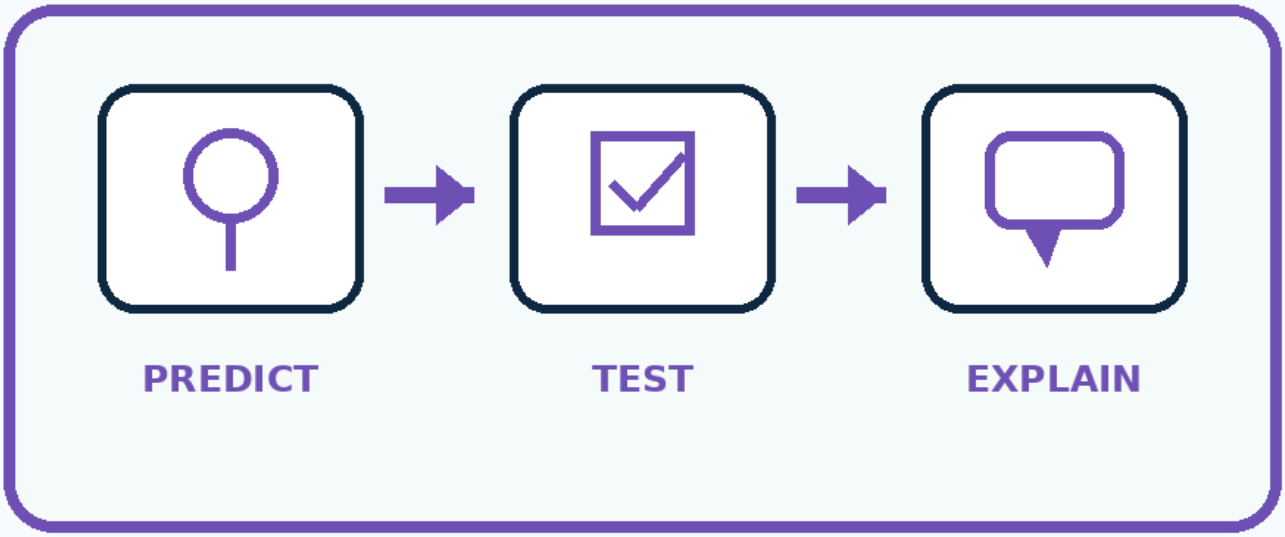
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EXPLORER LAB

One-Cause Improvement Cycle

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Revise one rule or **input** and rerun the same tests.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: audit the different equipment repair

> Compare U0/U1 against the equipment contract before proposing U2.

The independent catalogue and version changes

Equipment catalogue E lists E1 first with capacity 4 and E2 second with capacity 8. Valid whole group sizes are 1 to 10. Choose the first capacity at least the group size, or no match; invalid **input** needs clarification.

Old U0 uses strictly greater capacity and stops at the first fit. Proposed U1 includes equality but keeps searching and returns the last fit. Independently compare U0, U1 and the requirement for groups 8 and 3.

Identify the **repair** and **regression**, propose U2 and give one validation check. No booking occurs.

Keep your first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Audit a different proposed repair

> **My independent explanation, trace and limits**

Week 35 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Dim photos fail. You change the lighting and all decision rules together; performance improves. Can you tell which change helped? Design a clearer comparison.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

If the sorter has an **evidence**d defect, save a new version with one **evidence**-linked change and a before/after test table; otherwise retain the current tested version and state that no repair was needed.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Version A makes four errors. Version B makes one after lighting, labels and rules all change. What can you conclude, and how would you isolate the effect of lighting?

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Transfer evidence-linked repair to the sorter portfolio

> A repair is conditional on a located defect, not a compulsory version-number change.

Connect the two contexts and the no-change option

The connected sequence repairs the activity prototype only when evidence requires it. Transfer that method to your existing sorter portfolio here. A new version needs a justified change; if your sorter already meets the tested requirements, retain it and report that result. Keep sorter and activity regression records distinct, and do not change a candidate during its final evaluation.

This week’s sorter evidence

THIS WEEK'S CHECKPOINT If the sorter has an **evidence**d defect, save a new version with one **evidence**-linked change and a before/after test table; otherwise retain the current tested version and state that no repair was needed.

Maintain sorter **regression** tests and compare complete version traces, retaining new failures rather than hiding them.

Inspect the cause, intervention and retained checks

- What real classroom need does this serve?
- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?
- When should it say not sure or ask a person?

Inspect new failures as well as totals

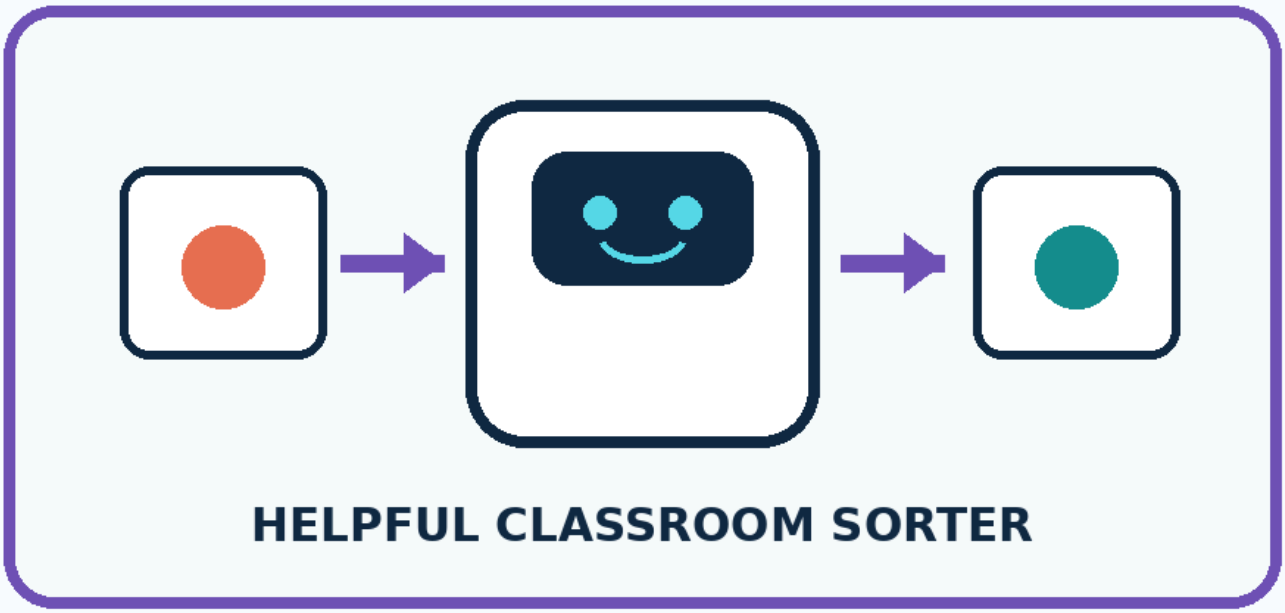
RESPONSIBLE DESIGN A higher pass count can hide a serious new failure; review every changed outcome.

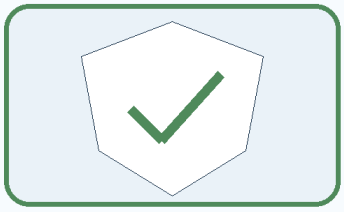
▼ Earlier wording, alternatives and picture transcripts

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: explain the repair, regression and limits

> Close your notes and point to the unchanged requirement behind your judgment.

Your exit explanation

- Close your notes. Explain this goal in your own words: **Repair** a located **defect** without changing the agreed requirements

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Choose the next regression check

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	Meaning	My example
root cause	earliest responsible reason for a failure	
controlled change	one targeted modification while other factors stay fixed	
retest	repeat the same case after a change	
regression	previously working behavior that becomes worse	

Terms for the connected explanation

Word	Meaning
Defect	A fault in the instructions or implementation that can cause incorrect behaviour
Repair	A change intended to correct a located defect
Change record	A note of what changed, why, and which versions and tests support it

Word	Meaning
Fixed candidate	A saved version kept unchanged while a specified evaluation is run

Repair a claim using a word

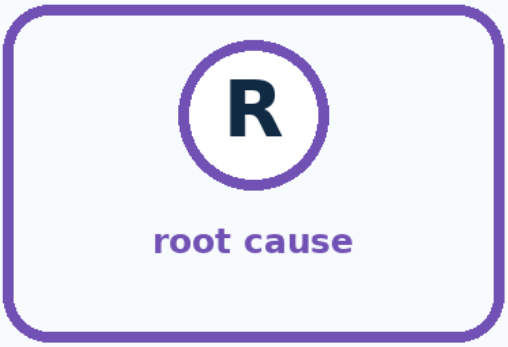
- CLAIM TO **REPAIR** Any change after a failure counts as improvement.
- Use one vocabulary word and one piece of **evidence** to **repair** it:

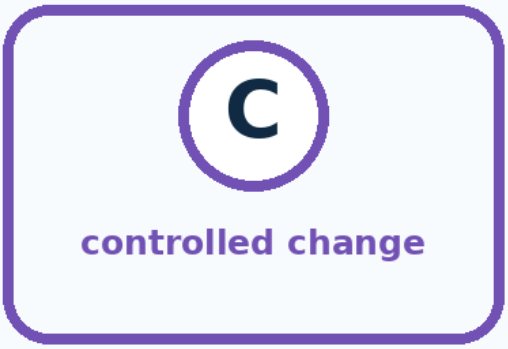
▼ Earlier wording, alternatives and picture transcripts

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ Choose a stronger thinking challenge

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

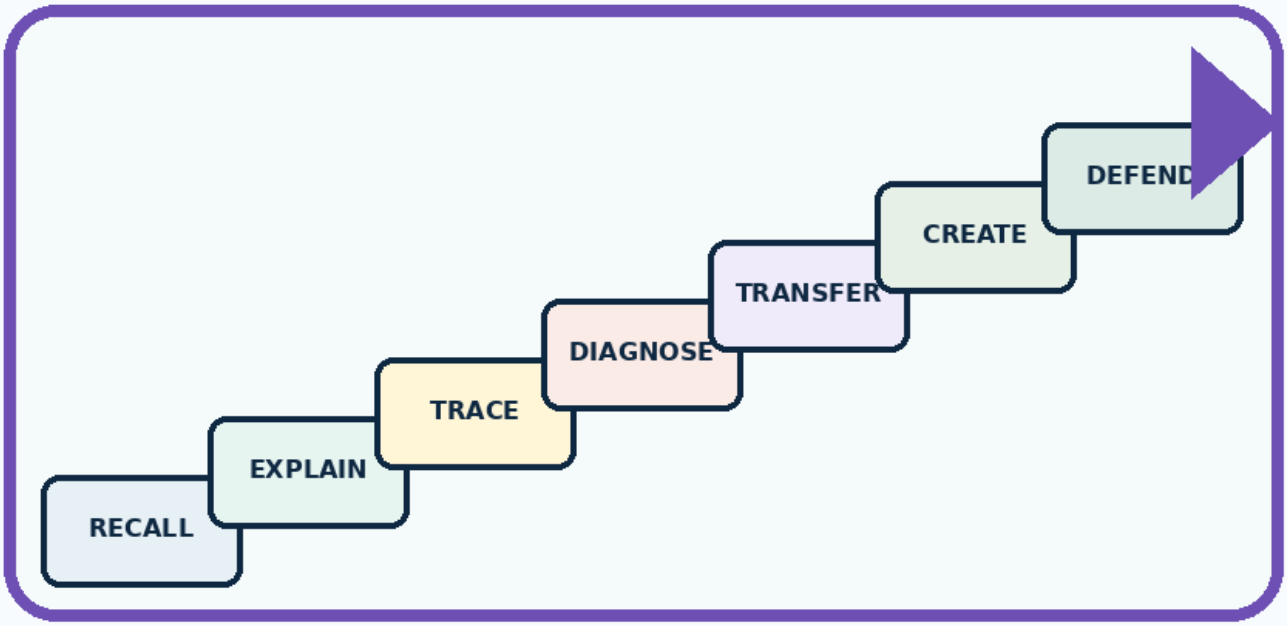
- RECALL Define **root cause**.
- EXPLAIN An improvement should target **evidence** from a failure, not random change.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE **Repair**: Any change after a failure counts as improvement.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternatives and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check your explanation**

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Audit a different proposed repair is the selected independent case.

Check the evidence and its limit

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Record support

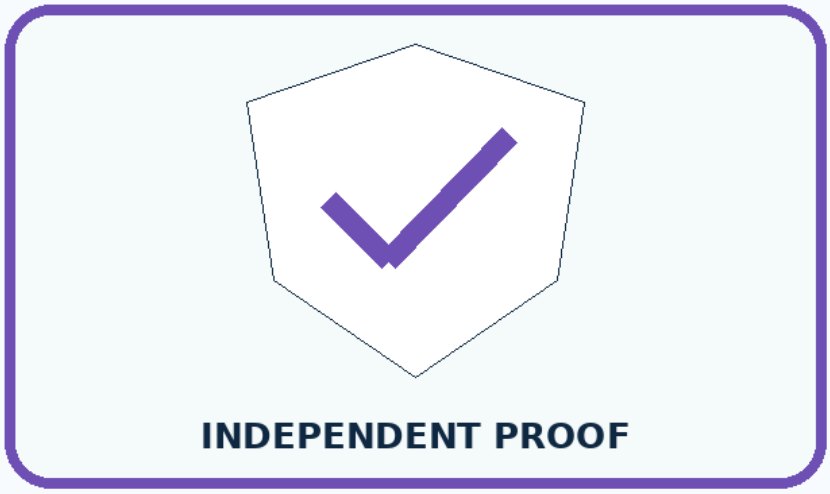
My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ Earlier wording, alternatives and picture transcripts

MASTERY

Checklist for explaining this week's idea

Use after practice. Audit a different proposed repair is the selected independent case.



EXPLANATION CHECK Locate the defect, preserve all other requirements, rerun earlier successes and retain versioned evidence before fixing a final candidate.

Independent **evidence**:

SELF-CHECK Name the **evidence**, suspected cause, single change, and predicted effect before editing. Rerun the full fixed suite, compare case by case, and keep the change only if benefits outweigh new failures.

NEXT BRIDGE Carry today's **evidence** habit into: Expo and Defense.

▼ **Extra practice: A fair comparison changes one cause**

REFERENCE · SECTION 13

Extra practice: A fair comparison changes one cause

> Try the question before opening its worked explanation.

Read the supplied case and question

The team changes lighting, labels and rules together; errors fall from four to one.

Can it identify the cause of improvement?

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

No. Several factors changed. Compare one planned change at a time with controlled conditions.

Evidence or stage	Value or result
Before	Four errors under recorded conditions
Confounded revision	Lighting, labels and rules all changed
Controlled revision	Change one planned factor and retest

▼ Earlier wording, alternatives and picture transcripts

Week 35 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

A fair comparison changes one cause

The team changes lighting, labels and rules together; errors fall from four to one.

Can it identify the cause of improvement?

No. Several factors changed. Compare one planned change at a time with controlled conditions.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Repair a located defect without changing the agreed requirements	Explain the deciding step, show a trace or test, and state one limit.
Check the repaired case and earlier successful behaviour	Explain the deciding step, show a trace or test, and state one limit.
Preserve versioned evidence and fix a candidate before final evaluation	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional quiet-activity chooser has a boundary bug and a proposed **repair**.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional quiet-activity chooser has a boundary bug and a proposed **repair**.
- Which labelled step matches this detail? U1 selects noisy B at 6; fixing equality introduced a new failure.
- 1. Bad revision
- 2. **Regression** case
- 3. Expected **repair**
- 4. Boundary failure
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Bad revision U1 selects noisy B at 6; fixing equality introduced a new failure. A **repair** must preserve the full contract, not just make one failing example pass.

The quiet condition and duration comparison are separate requirements. Changing only the inequality targets the recorded cause; removing the quiet check changes another behavior.

Save previously passing cases to reveal such **regression**s.

Changed-case question

- Changed-case practice: Use U2, which changes < to ≤ and keeps the quiet check. Which conclusion is supported by the supplied facts?
- 1. At 8 return A; at 6 return none.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed case**

After your attempt, compare with the supported explanation: At 8 return A; at 6 return none. Equality is **repair**ed while B remains ineligible because it is noisy.

Repair and save your work

Review Test Edge Cases and Fairness (ai-1/34). Goal: Choose tests just below, at and above a decision boundary. Short practice: Which single boundary run most directly detects the strict-comparison bug, and why?.

Return to this week and explain what you changed.

Review Build a Paper Prototype (ai-1/32). Goal: Build a runnable paper prototype with explicit instructions and visible state. Short practice: What should appear in the test log before and after the **repair**?.

Return to this week and explain what you changed.

Review Find a Real Problem (ai-1/29). Goal: Describe a specific user difficulty using supplied **evidence** without inventing its cause.

Short practice: Write a one-sentence project goal and say what **evidence** would count as a failure.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

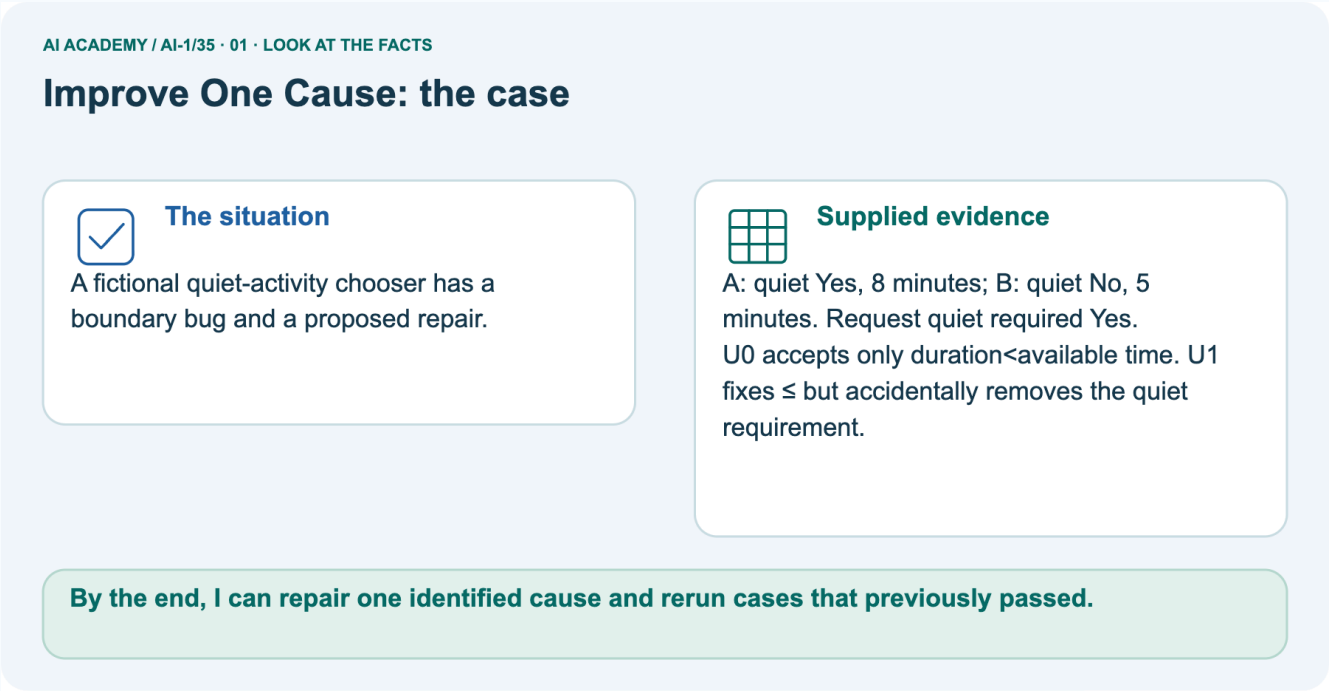
Keep your completed work

▼ Reference: the six-picture journey

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Improve One Cause: the case



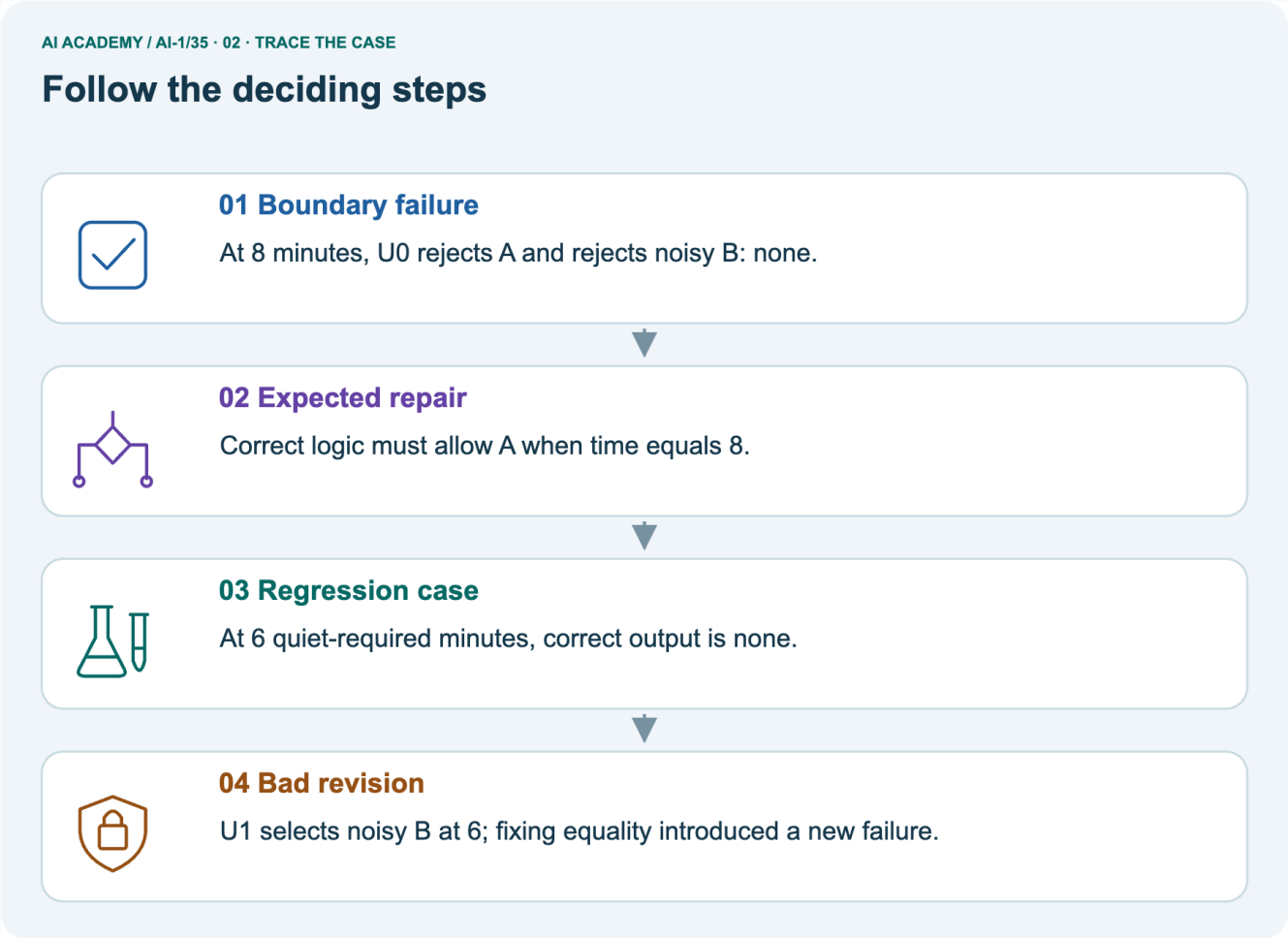
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional quiet-activity chooser has a boundary bug and a proposed **repair**. Supplied facts: A: quiet Yes, 8 minutes; B: quiet No, 5 minutes. Request quiet required Yes. U0 accepts only duration<available time.

U1 fixes ≤ but accidentally removes the quiet requirement.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

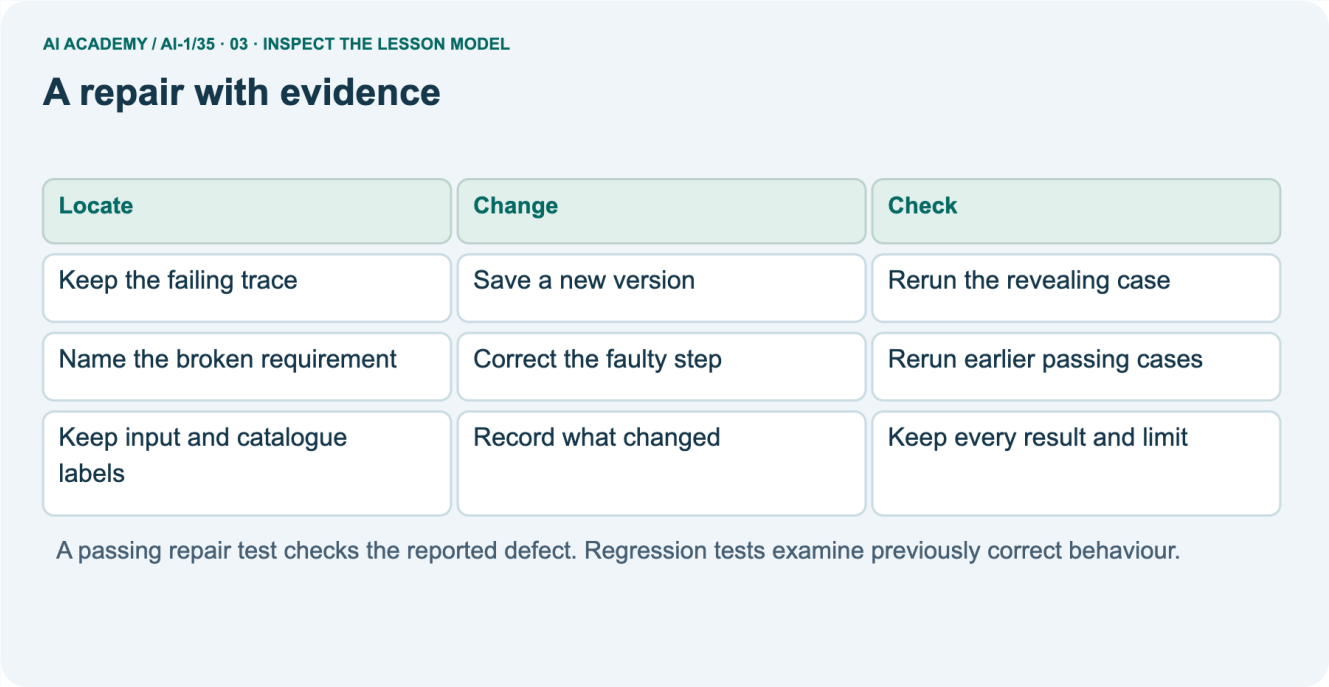


Follow the labelled picture with the matching explanation. Open the vector diagram

Boundary failure: At 8 minutes, U0 rejects A and rejects noisy B: none. → Expected **repair**: Correct logic must allow A when time equals 8. → **Regression** case: At 6 quiet-required minutes, correct **output** is none. → Bad revision: U1 selects noisy B at 6; fixing equality introduced a new failure.

Point to each numbered stage and explain which recorded **input** it uses.

03. A repair with evidence



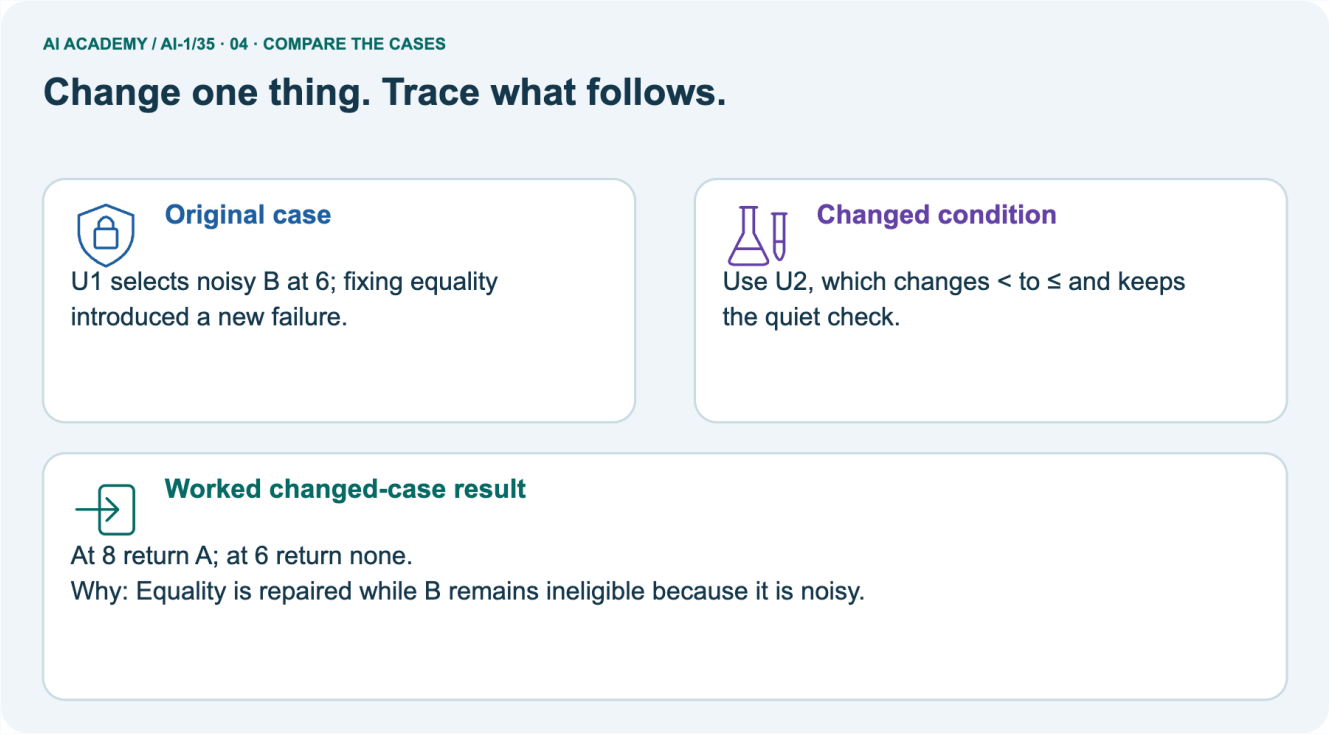
Follow the labelled picture with the matching explanation. Open the vector diagram

A **repair** with **evidence**. Keep the failing trace | Save a new version | Rerun the revealing case. Name the broken requirement | Correct the faulty step | Rerun earlier passing cases.

Keep **input** and catalogue labels | Record what changed | Keep every result and limit

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



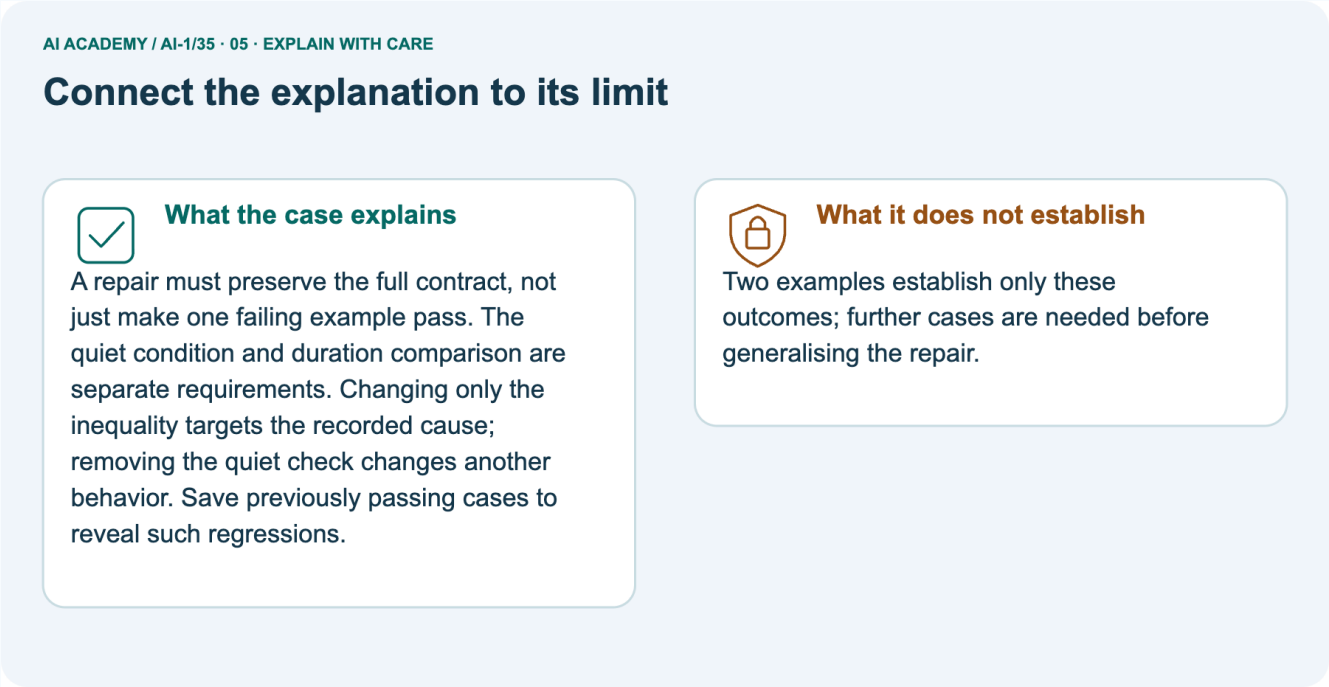
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: U1 selects noisy B at 6; fixing equality introduced a new failure. Changed condition: Use U2, which changes < to ≤ and keeps the quiet check. Expected result: At 8 return A; at 6 return none.

Reason: Equality is **repair**ed while B remains ineligible because it is noisy.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

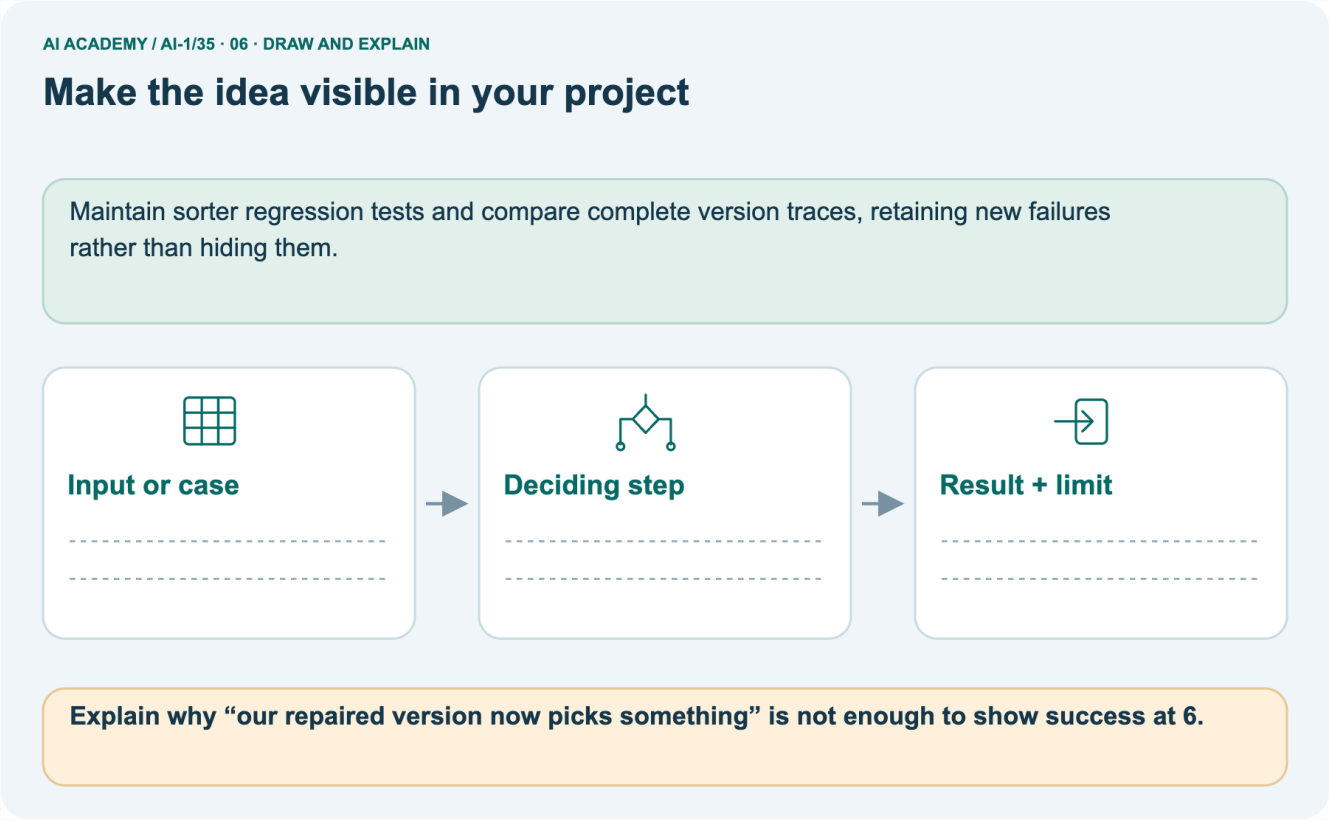
A **repair** must preserve the full contract, not just make one failing example pass. The quiet condition and duration comparison are separate requirements.

Changing only the inequality targets the recorded cause; removing the quiet check changes another behavior. Save previously passing cases to reveal such **regression**s.

Limit: Two examples establish only these outcomes; further cases are needed before generalising the **repair**.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Maintain sorter **regression** tests and compare complete version traces, retaining new failures rather than hiding them. Draw your **input**, deciding step, result and limit.

Supported question: Explain why “our **repair**ed version now picks something” is not enough to show success at 6.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Improve One Cause: the case

02. Follow the deciding steps

03. A repair with evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.